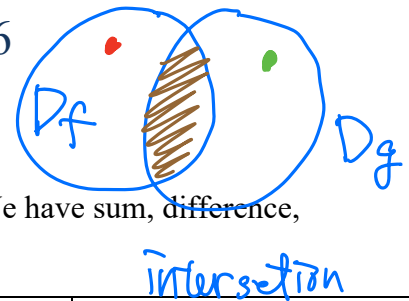


Ch5. Operations on Functions

1. Complete the definition of the Algebra of Functions:

Let $f(x)$ and $g(x)$ be two functions with the domain D_f and D_g , respectively. We have sum, difference, product, and quotient of functions:



The Algebra of functions	Notation	Definition	Domain
Sum	$(f+g)(x) := f(x) + g(x)$		$D_{f+g} = D_f \cap D_g$
Difference	$(f-g)(x) := f(x) - g(x)$		$D_{f-g} = D_f \cap D_g$
Product	$(fg)(x) := f(x) \cdot g(x)$		$D_{f \cdot g} = D_f \cap D_g$
Quotient	$\left(\frac{f}{g}\right)(x) := \frac{f(x)}{g(x)}$, provided $g(x) \neq 0$		$D_{\frac{f}{g}} = D_f \cap D_g$ but $g(x) \neq 0$

Here, $D_f \cap D_g = \{x \mid \underline{x \text{ is from } D_f \text{ and } x \text{ is from } D_g}\}$

$$D_f = (-\infty, \infty) \quad D_g = (-\infty, \infty)$$

2. Let $f(x) = x^2 + 5x + 6$ and $g(x) = x + 2$. Find the following functions and state their domains.

$$(f+g)(x) = f(x) + g(x) = x^2 + 5x + 6 + x + 2 = x^2 + 6x + 8. \quad D_{f+g} = (-\infty, \infty) \text{ all real numbers}$$

$$(f-g)(x) = f(x) - g(x) = x^2 + 5x + 6 - (x + 2) = x^2 + 5x + 6 - x - 2 = x^2 + 4x + 4$$

$$D_{f-g} = (-\infty, \infty)$$

$$(f \cdot g)(x) = f(x) \cdot g(x) = x^3 + 7x^2 + 16x + 12. \quad D_{f \cdot g} = (-\infty, \infty) \text{ all real numbers}$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^2 + 5x + 6}{x + 2}$$

$$= \{x \mid x \in (-\infty, \infty)\}$$

$$= \{x \mid x \in \mathbb{R}\}$$

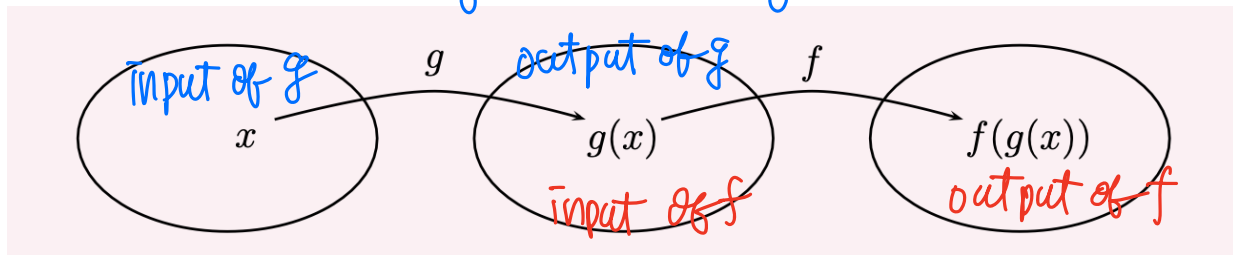
$$= \frac{(x+3)\cancel{(x+2)}}{\cancel{(x+2)}} = x+3 \text{ but } x \neq -2$$

$$D_{\frac{f}{g}} = \{x \mid x \in (-\infty, -2) \cup (-2, \infty)\} = \{x \mid x \in \mathbb{R}, \text{ but } x \neq -2\}$$

3. Complete the definition of the Composition of Functions:

Let $f(x)$ and $g(x)$ be two functions. The composition of the function f with g is denoted by

$(f \circ g)(x)$ and is defined by the equation $(f \circ g)(x) := f(g(x))$.



The domain of the composition of the function $f \circ g$ is the set of all x such that x is the input of $g(x)$ and output of $g(x)$ is the domain of $f(x)$.

The notation of the domain of the composition of the function $f \circ g$ is

$D_{f \circ g} = \{x \mid x \text{ is from } D_g \text{ and output of } g(x) \text{ is from } D_f\}$

4. Are $f(g(x))$ and $g(f(x))$ the same functions?

$f(p) = p \cdot 0.8$ $g(p) = p - 500$ **NO**
 $g(f(p)) = f(p) - 500 = p \cdot 0.8 - 500$

$f(g(p)) = g(p) \cdot 0.8 = (p - 500) \cdot 0.8$

5. Find $(f \circ g)(x)$ for the following functions and state their domains.

a) $f(x) = x^2 + 2$ and $g(x) = x - 3$

$f(g(x)) = f(x - 3) = (x - 3)^2 + 2$
 $= x^2 - 6x + 9 + 2 = x^2 - 6x + 11$
 $D_{f \circ g} = (-\infty, \infty) = \{x \mid x \in \mathbb{R}\}$

b) $f(x) = \frac{2}{x-3}$ and $g(x) = x^2 + 2x$